

Paper 2 Practice (Mark Scheme)

Part	Answer	Marks	Partial Marks / Guidance
1(a)	10	2	M1 for $\sqrt{8^2 + 6^2}$
1(b)	5	1	B1 for $\frac{1}{2} \times 10$
1(c)	60°	2	M1 for $\tan(\angle VYM) = \frac{5\sqrt{3}}{5}$
2(a)	Show proof	3	M1 for $\frac{30}{x} - \frac{30}{x+5} = \frac{1}{2}$
2			M1 for $30(x+5) - 30x = \frac{1}{2}x(x+5)$
2			A1 for establishing $x^2 + 5x - 300 = 0$ with no errors
2(b)	2 hours	3	M1 for $(x+20)(x-15) = 0$ or correct substitution into formula
2			A1 for $x = 15$ (rejecting $x = -20$)
2			A1 for time = $\frac{30}{15} = 2$
3(a)(i)	0.2 or $\frac{1}{5}$	1	B1
3(a)(ii)	$\frac{1}{9}$	2	M1 for $(-27)^{\frac{1}{3}} = -3$ or $\frac{1}{(-27)^{\frac{2}{3}}}$ or $\frac{1}{9}$
3(b)	$\frac{2a^8}{27}$	2	M1 for cubing brackets to get $\frac{8a^{15}}{27}$
4(a)	1	1	B1
4(b)	$\frac{1}{16}$	1	B1
4(c)	$\frac{3}{8}$	1	B1
5(a)	$\frac{2x}{\frac{x}{60} + \frac{x}{90}} = 72$	3	M1 for setting total distance as $2x$, M1 for total time expression $\frac{x}{60} + \frac{x}{90} = \frac{5x}{180} = \frac{x}{36}$, A1 for $\frac{2x}{\frac{x}{36}} = 72$

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5(b)	120	2	M1 for total distance = $72 \times 3\frac{1}{3} = 72 \times \frac{10}{3} = 240$ km, then $\frac{240}{2}$
6(a)	33°	3	B1 for $\angle ACD = 90^\circ$ (Angle in a semicircle) M1 for setting $\angle BCA = \angle CAD = x$ (Alternate angles) and forming $x + (x + 90^\circ) + 24^\circ = 180^\circ$ A1 for 33°
6(b)	123°	2	M1 for $\angle BCD = \angle BCA + \angle ACD = 33^\circ + 90^\circ$ A1 for 123°
6(c)	66°	1	B1 for 66°
6(d)	Isosceles trapezium	2	B1 for Isosceles trapezium B1 for stating it has one pair of parallel sides and non-parallel sides are equal chords (derived from symmetry/cyclic properties)
7(a)	38°	1	B1 for 38° (Isosceles triangle property)
7(b)	38°	1	B1 for 38° (Angles subtended by the same arc)
7(c)	62°	3	M1 for $\angle BAC = \angle BDC = 38^\circ$ M1 for $\angle BAD = 38^\circ + 38^\circ = 76^\circ$ A1 for evaluating $180^\circ - 42^\circ - 76^\circ = 62^\circ$
7(d)	Proof	3	M1 for $\angle PAB = \angle PDC$ (Angles in same segment) M1 for $\angle PBA = \angle PCD$ (Angles in same segment) or vertically opposite angles at P A1 for concluding AAA similarity explicitly
8(a)	1000	3	M1 for writing individual values to 1 sig fig as 40, 5, and 0.2, M1 for $\frac{40 \times 5}{0.2}$ or $\frac{200}{0.2}$, A1 for 1000.
9(a)	$6x^3 - 13x^2 - 49x + 40$	3	M1 for expanding two brackets correctly to get $6x^2 + 11x - 10$ or $2x^2 - 3x - 20$ or $3x^2 - 14x + 8$
9			M1 for full expansion of all terms with at most two errors in expansion or signs

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9(b)(i)	$6ab(2a - 3b)(2a + 3b)$	2	M1 for $6ab(4a^2 - 9b^2)$ or correct partial factorisation with higher common factor remaining
9(b)(ii)	$2(2x - 3)(x - 2)(x + 2)$	3	M1 for grouping to get $2x^2(2x - 3) - 8(2x - 3)$ or $4x(x^2 - 4) - 6(x^2 - 4)$
9			M1 for factorised form containing a difference of two squares like $(2x^2 - 8)(2x - 3)$ or $(4x - 6)(x^2 - 4)$
10(a)	$50000 \times 0.8^2 = 32000$	2	M1 for 50000×0.8^2 or $50000 \times \left(\frac{4}{5}\right)^2$ or 40000 seen
10(b)	25600 EUR	2	M1 for 32000×0.8 or 50000×0.8^3
11(a)	$\frac{1}{8}$	1	B1 for correct fraction simplified completely
11(b)	2.6	1	B1
12(a)	9 litres per minute	2	M1 for $\frac{150 \times 60}{1000}$ or equivalent
12(b)	Show that height increases by 3.6 mm	3	M1 for converting volume per hour: $9 \times 60 = 540$ litres or 0.54 m^3
12			M1 for height = $\frac{\text{volume}}{\text{area}} = \frac{0.54}{2.5} = 0.216 \text{ m}$
12			A1 for converting 0.216 m to 216 mm and dividing by 60 to get 3.6 mm or equivalent complete proof
13(a)	$67.5\%, \frac{17}{25}, 0.682, \left(\frac{1}{8}\right)^{-1} \times 10^{-1}$		M1 for evaluating individual terms to equivalent forms: 0.68, 0.675, 0.682, 0.8
13(b)	$=$ or $\geq$ or $\leq$	1	B1 for verifying both sides are structurally equivalent to 0.035
14(a)	605 grams	3	M2 for $500 \times \left(\frac{110}{100}\right)^2$ or $500 \times 1.1 \times 1.1$ or M1 for $500 \times 1.1 = 550$
14(b)	121%	1	B1 for 121% or their $\frac{605}{500} \times 100$
15(a)	‘9’ and ‘15’	3	M1 for setting up equation ‘ $\frac{1}{2}(3k + 5k) \times 8 = 96$ ’

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15			M1 for simplifying to ' $32k = 96$ ' or ' $k = 3$ '
15(b)	$'24 + 2\sqrt{73}'$	4	M1 for identifying the base of the right-angled side triangle is ' $\frac{15-9}{2}$ ', or ' 3 '
15			M1 for formulating Pythagorean theorem ' $\sqrt{3^2 + 8^2}$ '
15			M1 for exact slant side ' $\sqrt{73}$ '
16(a)	-9	2	M1 for $2 \times 9 - 9 \times 3$ or $18 - 27$
16(b)	-105	2	M1 for $10 \times 2 - 125$ or $20 - 125$
17(a)	£1200	2	M1 for $\frac{4}{12} \times 3600$ oe
17(b)	Manager C, loss of £100	3	M1 for finding new shares: $A = 800, B = 1200, C = 1600$, M1 for comparing old shares ($A = 900, B = 1200, C = 1500$) to identify A loses £100 and C gains £100, A1 for Manager A with £100 loss
18*	$x = \frac{2}{3}$ or $x = 4$	3	M2 for $(3x - 2)(x - 4) = 0$
18			or M1 for $3x(x - 4) - 2(x - 4) = 0$ or correct split of middle term.
19(a)	3.6×10^5	2	M1 for $43 \times 10^4 - 7 \times 10^4$ or $430\,000 - 70\,000$ or $360\,000$
20(a)	$4\sqrt{2}$	1	
20(b)	$2\sqrt{6}$	2	M1 for $\frac{12\sqrt{6}}{6}$
21(a)	4	2	M1 for $5^2 - 3^2$ or $\sqrt{25 - 9}$
21(b)	4 cm	1	
21(c)	12 cm^2	2	M1 for $2 \times \left(\frac{1}{2} \times 3 \times 4\right)$ or $\frac{1}{2} \times 3 \times 4$
22(a)	5.25	1	B1 for 5.25
22(b)	0.024	1	B1 for 0.024

Part	Answer	Marks	Partial Marks / Guidance
23(a)	10:45 p.m.	1	B1 for correct time format with p.m.
23(b)(i)	19 20	2	M1 for adding 9 hours and 11 hours 35 minutes to 22 45 or equivalent
23(b)(ii)	Wednesday	1	B1 for Wednesday
24(a)	$2^2 \times 3^3 \times 5$	2	M1 for any correct decomposition process (e.g., factor tree or repeated division) showing at least 3 correct prime factors
24(b)	15	2	M1 for identifying that the powers of 3 and 5 must be even, so $k = 3 \times 5$
24(c)	$\frac{1}{15}$	1	B1 ft from their value of k